

ILHA BYRNE

PhD candidate in ecology & evolution at the University of Queensland

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Education

2014 – 2015: Bachelor of Science (Honours), *First Class Honours, First in Course Award*

School of Biological Sciences, University of Queensland

Thesis: “DNA metabarcoding as a tool to characterise the spatio-temporal distribution of planktonic echinoderm larvae.”

2017 – 2020: Bachelor of Science, *Faculty Graduate of the Year, University Medal*

University of Queensland

GPA: 6.9/7

Employment History

2024 – 2025: Casual Research Assistant, Faculty of Science and Engineering, Southern Cross University

Conduct molecular lab work for Dr Ordóñez Álvarez associated with a CCA project aimed at resolving species identities and phylogenetic relationships.

2023 – present: Casual Research Assistant, School of the Environment, University of Queensland

Conduct molecular lab work and field work for the Marine Spatial Ecology Lab, The Tibbetts Lab and the Riginos Research Group.

2022 – 2024: Casual Team Leader, Reef Check Australia

Organised and led community events and underwater surveys.

2021 – 2023: Full-time Research Assistant, School of Biological Sciences, University of Queensland

Facilitated eDNA lab work, marine SCUBA fieldwork, stakeholder workshops and budget keeping for the Sustainable Urban Seascapes Moreton Bay. Conducted coral collection fieldwork and genomic lab work for the Reef Restoration and Adaptation program.

2019 – 2021: Casual Administration Assistant, Faculty of Science, The University of Queensland

Assisted with student enquires and general office administration and communications.

2018 – present: Casual Academic, Faculty of Science, The University of Queensland

Tutor and demonstrate undergraduate courses in various roles (see Teaching below).

2018 – 2024: Casual Engagement Assistant, Faculty of Science, The University of Queensland

Facilitated and developed workshops for high school students in science, as well as on-campus STEM and recruitment events.

2016 – 2017: Divemaster, Blue World Diving, Kinasi Lodge

Led dives, assisted with teaching PADI SCUBA courses, and facilitated the day to day running of the dive shop.

Teaching

2023 – present: International Programs Tutor, The University of Queensland

University of California Study Abroad Program (3 semesters)

Stanford University Study Abroad Program (2 semesters)

2017 – present: Demonstrator, The University of Queensland

‘GEOS1100 – Environment & Society’ (2 semesters)

‘BIOL2204 – Zoology’ (2 semesters)

‘MARS1001 – Marine Science’ (2 semesters)

2018 – 2020: Peer Assisted Study Session Tutor, School of Earth and Environmental Science, The University of Queensland

Provided weekly, university-funded academic tutoring for undergraduate students enrolled in ‘ERTH1000 – Planet Earth: The Big Picture’.

2018 – 2020: Science Learning Centre Tutor, Faculty of Science, The University of Queensland

Provided weekly, university-funded academic tutoring and mentoring for undergraduate students enrolled in Biology, Chemistry, Genetics and Statistics courses at the University of Queensland.

Awards and Prizes

2025 3MT School of Environment Runner-up, Faculty of Science Finalist, The University of Queensland

2023 Outstanding Service and Citizenship, School of Biological Sciences, The University of Queensland

2022 Best Student Presentation, Australian Coral Reef Society Conference

2022 University Medal for Outstanding Achievement, The University of Queensland

2022 Honours Prize (Top student), School of Biological Sciences, The University of Queensland

2021 Future Leaders’ Award, The University of Queensland

2021 Employability Award, The University of Queensland

2019 Genetics Prize (Top student in second year), The University of Queensland

2019 Gordon Griff Vertebrate Zoology Prize (Top student in second year), The University of Queensland

2017 Dean’s List for Academic Excellence, Faculty of Science, University of Queensland (**2017 – 2021**)

Grants and Research Funding

2025 Holsworth Student Research Endowment, Ecological Society of Australia (**\$8,000 AUD**)

2024 Student Travel Award, The Australian Coral Reef Society (**\$500 AUD**)

2022 Lizard Island Research Station Award, The Australian Coral Reef Society (**\$5,000 AUD**)

Competitive Scholarships

2023 Global Change Scholars Program, The University of Queensland and CSIRO (**2023 – 2024**)

2023 ELEVATE Boosting Women in STEM Scholarship (**\$80,000 AUD, 2023 – 2026**)

2023 Research Training Program Scholarship (**2023 – 2027**)

2019 Summer Research Scholarship, School of Biological Sciences, University of Queensland

2018 Summer Research Scholarship, School of Biological Sciences, University of Queensland

Publications

See a list of my publications on my [Research Gate](#) and [Google Scholar](#) pages.

Conference Presentations

2025: Byrne I et al. 'Causal modelling reveals lower coral genetic diversity in more heterogenous reef environments'. International Congress on Conservation Biology. 15-20 June, Brisbane, Australia.

2024: Byrne I et al. 'Estimates of genetic diversity and differentiation in the Caribbean coral *Agaricia tenuifolia*'. Australian Coral Reef Society Conference. 6-8 May, Perth, Australia.

2022: Byrne I et al. 'Monitoring the benthic marine communities in Quandamooka, Moreton Bay'. Australian Coral Reef Society Conference. 25-27 November, Brisbane, Australia.

2022: Roelfsema C et al. 'Citizen Science: The Mooloolaba Ecological Assessment and Mapping Project'. Australian Coral Reef Society Conference. 25-27 November, Brisbane, Australia.

2022: Byrne I et al. 'DNA metabarcoding as a tool to detect and monitor echinoderm larvae'. Australian Marine Science Association Conference. 7-11 August, Cairns, Australia.

Conference Posters

2025: Byrne I et al. 'Genomic evidence of cryptic species and habitat specialization in the brooding coral *Seriatopora hystrix* along the Great Barrier Reef'. Congress of the European Society for Evolutionary Biology. 17-22 August, Barcelona, Spain.

Public Talks

2025: 'Causal modelling reveals lower coral genetic diversity in more heterogenous reef environments'. **Centre for Marine Science Seminar.**

2024: 'Navigating the publication process as a postgraduate student'. **Centre for Marine Science Seminar.**

2023: 'eDNA monitoring of the benthic marine communities in Quandamooka, Moreton Bay'. **Reef Check Australia Public Webinar [video].**

Professional Service

Secretary, Society for Conservation Biology (South East Queensland) (2024 – present)

HDR Chair, Centre for Marine Science Committee (2023 – 2025)

2025 UQ-CNRS West Pacific Marine Biology Symposium Organising Committee

HDR representative, Australian Marine Science Association Committee (South East Queensland branch, 2023 – 2025)

2023 HDR Conference Organising Committee

Science Outreach

Curious Minds Coach: Mentored two high-school students who are aspiring women in STEM (2024 – present).

Wonder of Science Program: Led and supported STEM activities for school kids across QLD, including in regional and remote communities. The WoS Program is an initiative of the office of the Queensland Chief Scientist (2023 – 2025).

Science Engagement Officer: Facilitated in-classroom and outdoor STEM experiences for high school students (2018 – 2024).

Coast to Coral Seminar Series Coordinator: Organised and hosted the Reef Check Australia monthly online seminar series (2022 – 2024).

Reef Check Australia Ambassador: Represented Reef Check at markets, festivals, and invited community events (2018 – present).

Center for Marine Science Volunteer: Participated in and organised shore walks, roadshows, and talk fests (2018 – 2021).

Science Peer Mentor: Faculty of Science mentor to first-year undergraduate students at UQ (2018 – 2020).

Undergraduate Leadership

Women in Ocean Science University Ambassador: Organised events for undergraduate women and girls studying marine and ocean science at UQ (2023).

Women in Science Association Postgraduate Officer: Helped to organise events and liaise with postgraduate members (2023).

Marine Society UQ SEQUEST Officer: Facilitated and organised the SEQUEST project which aimed to collect long-term data on marine invertebrate communities in Moreton Bay (2021).

Marine Society UQ Treasurer: Served as the treasurer of the marine society and managed all financial issues (2020).

Marine Society UQ Co-founder and Vice President: Established the Marine Society at UQ and served as its first vice president (2019 – 2020).

Volunteer Experience

Reef Check Australia Surveyor: Conducted in-water reef health surveys and clean-ups (2018 – present).

Earth Observation Centre Field Volunteer: Assisted with in-water surveys designed to ground-truth satellite observations of coral reef benthic habitats.

Leeds University Field Volunteer: Conducted underwater fieldwork on the topicalization of reefs in northern New South Wales and Southern Queensland (2023, 2024).

Coral Reef Ecosystems Laboratory Field Volunteer: Assisted with underwater coral calcification rate experiments at Lizard Island Research Station (2022).

Mooloolaba Ecological Assessment and Mapping: (2020 – 2022).

Bush Heritage Australia Volunteer: Assisted with collecting citizen-science data on intertidal marine organisms at Heron Island Research Station and conducted lantana removal and land rehabilitation at Avoid Island (2018 – 2020).

Laboratory Assistant: Volunteered with the Franklin, Cheney and Riginos research groups and assisted postgraduate students with their experiments and lab work (2018 – 2020).

In the media

Check out my science communication page on Instagram: [@biologist.byrne](#)

Decrypting the early-life stage secrets of marine species, Centre for Marine Science [article]

Sustainable Urban Seascapes Moreton Bay, Centre for Marine Science [article]

EcoRRAP, Reef Restoration Adaptation Program [video]

Let's chat about coral reefs, Centre for Marine Science [video]

Meet Ilha – she's studying Marine Biology at UQ, The University of Queensland: [video]

Professional Qualifications

2024 Coxswain Grade 1 Near Coastal (Paradigm Training)

2024 ADAS Part 1R Scientific Diver (The University of Tasmania)

2019 Recreational Boat License (ABC boating)

2016 PADI Divemaster (Big Blu Dive Centre)

Society memberships

Women in Technology (2023 – present)

Australian Coral Reef Society (2022 – present)

Australian Marine Sciences Association (2022 – present)

Marine Society UQ (2019 – 2024)

Women in STEM Association UQ (2017 – 2023)

Other Courses and Training

2025 Population genomic inference from low-coverage whole-genome sequencing data (Physalia)

2025 Summer School on Software and Statistical Methods for Population Genomics

2025 Workshop on Speciation and Population Genomics (Evomics)

2024 Introduction to Genetics and Evolution (Duke University)

2024 Bayesian Statistics: From Concept to Data Analysis (University of California, Santa Cruz)

2024 Advanced Spatial Statistics (Physalia)

2024 HLTAID009/015 Provide Cardiopulmonary Resuscitation and Oxygen Therapy (UQ Skills)

2023 HLTAID010/011 Provide First Aid (UQ Skills)

2022 Advanced Spatial Mapping in R (The University of Queensland)

2021 Introduction to Unix (The Queensland Cyber Infrastructure Foundation)

2021 Introduction to Python (The Queensland Cyber Infrastructure Foundation)

2020 Intermediate R course (The University of Queensland)

2019 R for Beginners (The University of Queensland)

Academic referees

Dr Cynthia Riginos, Professor, The University of Queensland, c.riginos@uq.edu.au

Dr Iva Popovic, Postdoctoral research fellow, The University of Queensland, i.popovic1@uq.edu.au

Dr Gurion Ang, Teaching-focussed academic, The University of Queensland, gurion.ang@uq.edu.au